

Programming Fundamentals 2

Topic 04 - Tutorial Sheet 4.1

CRUD (Create, Read, Update, Delete)

Section 1: Understanding CRUD

Question 1.1: Short Answers

- a) What does CRUD stand for?
 - b) Which CRUD operations were already implemented in Shop V3.0?
 - c) Which CRUD operations were missing in Shop V3.0?
 - d) Which CRUD operation(s) modify existing data?
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Section 2: Recap – Shop V3.0 Structure

Question 2.1: Classes and Responsibilities

- a) Write the instance field declaration used in `Store` to store products.
- b) Write the constructor for `Store` that initialises this collection.

```
public class Store {  
    // Your code here  
}
```

Section 3: CRUD and the Menu (Driver.java)

Question 3.1: Switch Statement

Write a `switch` statement fragment that handles the following menu options:

- 1 → Create product
- 2 → Read products
- 3 → Update product
- 4 → Delete product

```
switch(option) {  
    // Your code here  
}
```

Section 4: Delete (D) – Validation Methods

Question 4.1: productsExist Method

Write a method in `Store` that returns `true` if there is at least one product stored.

```
public boolean productsExist() {  
    // Your code here  
}
```

Question 4.2: Index Validation

Complete the method so that it checks whether an index is valid.

```
public boolean isValidIndex(int index) {  
    // index must be >= 0 and < products.size()  
    // Your code here  
}
```

Section 5: Delete (D) – Store Logic

Question 5.1: deleteProduct Method

Write the full method that deletes and returns a product if the index is valid. Return `null` otherwise.

```
public Product deleteProduct(int indexToDelete) {  
    // Your code here  
}
```

Section 6: Delete (D) – Driver Logic

Question 6.1: Driver Delete Case

Write the code for **case 4** in the `switch` statement that:

1. Checks products exist
2. Asks the user for an index
3. Attempts to delete the product
4. Prints a success or failure message

```
case 4 ->
    // Your code here
```

Section 7: Update (U) – Store Logic

Question 7.1: updateProduct Method

Write a method in `Store` that updates a product at a given index. Return `true` if the update succeeds, `false` otherwise.

```
public boolean updateProduct(int index, Product updatedProduct) {
    // Your code here
}
```

Section 8: Update (U) – Driver Logic

Question 8.1: Driver Update Case

Write the code for **case 3** in the `switch` statement that:

1. Checks products exist
2. Asks the user for an index
3. Reads in updated product details
4. Calls `updateProduct`
5. Prints a suitable message

```
case 3 ->
    // Your code here
```

Section 9: Read (R) – Code Practice

Question 9.1: listProducts Method

Write a method in `Store` that returns a formatted `String` containing all products with their index numbers.

```
public String listProducts() {
    // Your code here
}
```

Section 10: Create (C) – Code Practice

Question 10.1: addProduct Method

Write a method in `Store` that adds a `Product` to the `ArrayList`.

```
public void addProduct(Product product) {  
    // Your code here  
}
```

Section 11: Debugging

Question 11.1: Runtime Error

What is wrong with the following code?

```
products.remove(products.size());
```

Explain **why** it fails and how to fix it.

Question 11.2: Logic Error

Why is the following update logic unsafe?

```
products.set(index, updatedProduct);
```

Section 12: Extension (Optional)

Question 12.1: Safer Update

Rewrite the update logic so that it is safe and cannot cause a runtime error.
